University of Minho
School of Engineering

Intelligent Systems Associate Laboratory



Dados e Aprendizagem Automática

Linear and Logistic Regression

DAA 2026/2027

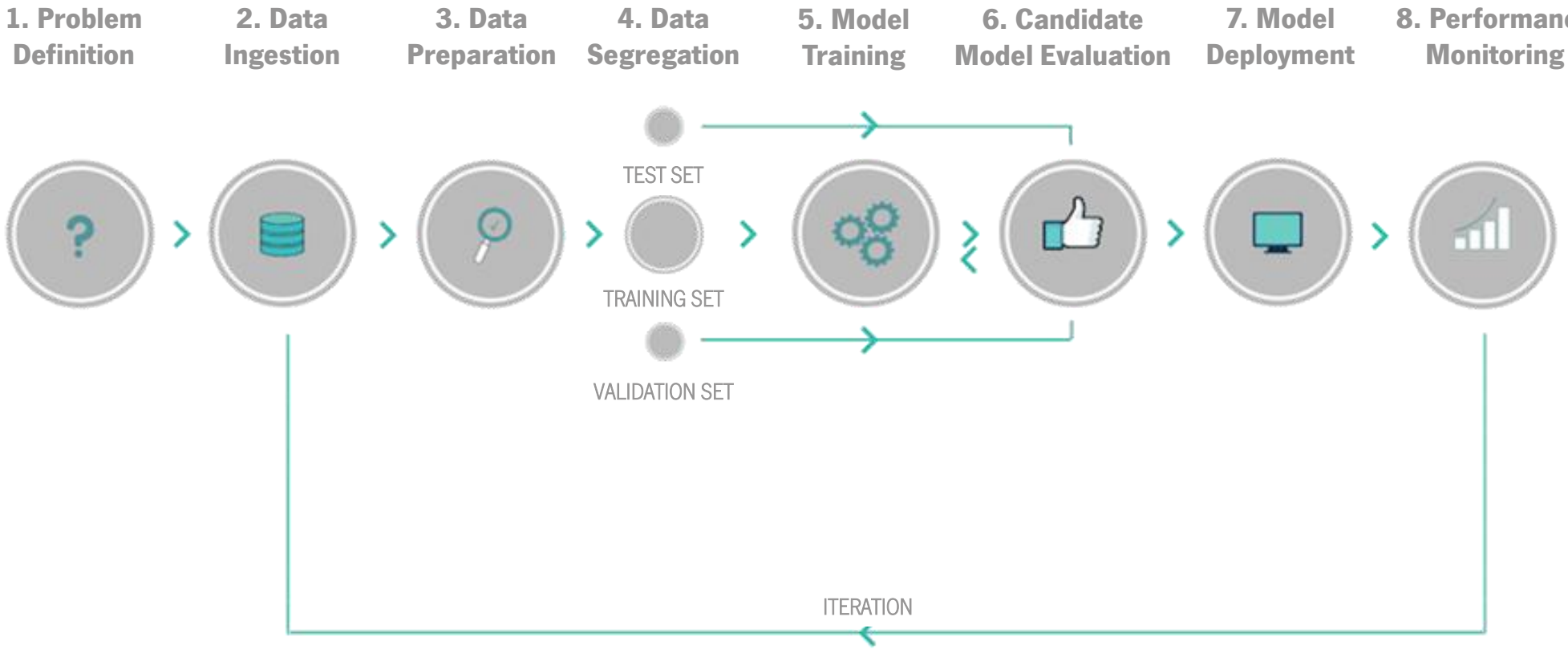
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Part IV

Contents

- Linear Regression
- Logistic Regression
- Hands On

Supervised Learning



We will explore the **Supervised Learning** algorithms.

The choice of the algorithm to implement depends on the type of data and the context provided. If we have labelled data or know the desired output in advance, we use the supervised learning paradigm.

Linear Regression

The Problem and the Data

Problem: development of a Machine Learning model capable of **predicting house prices** in regions across the USA.

Approach: **Linear Regression** approach.

Dataset: information about houses in regions of the United States, including:

- ***Avg. Area Income***: average income of residents in the city where the house is located;
- ***Avg. Area House Age***: average age of houses in the same city;
- ***Avg. Area Number of Rooms***: average number of rooms in houses in the same city;
- ***Avg. Area Number of Bedrooms***: average number of bedrooms in houses in the same city;
- ***Area Population***: population of the city where the house is located;
- ***Price***: the price at which the house was sold;
- ***Address***: the address of the house.

Exploratory Data Analysis


```
USAhousing = pd.read_csv('USA_Housing.csv')
```

```
USAhousing.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype  
---  ------                       --------------  -----  
 0   Avg. Area Income             5000 non-null   float64
 1   Avg. Area House Age          5000 non-null   float64
 2   Avg. Area Number of Rooms    5000 non-null   float64
 3   Avg. Area Number of Bedrooms 5000 non-null   float64
 4   Area Population              5000 non-null   float64
 5   Price                        5000 non-null   float64
 6   Address                      5000 non-null   object 
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

Exploratory Data Analysis


```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object 
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

```
USAhousing.columns
```

```
Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
       'Avg. Area Number of Bedrooms', 'Area Population', 'Price', 'Address'],
      dtype='object')
```

Exploratory Data Analysis

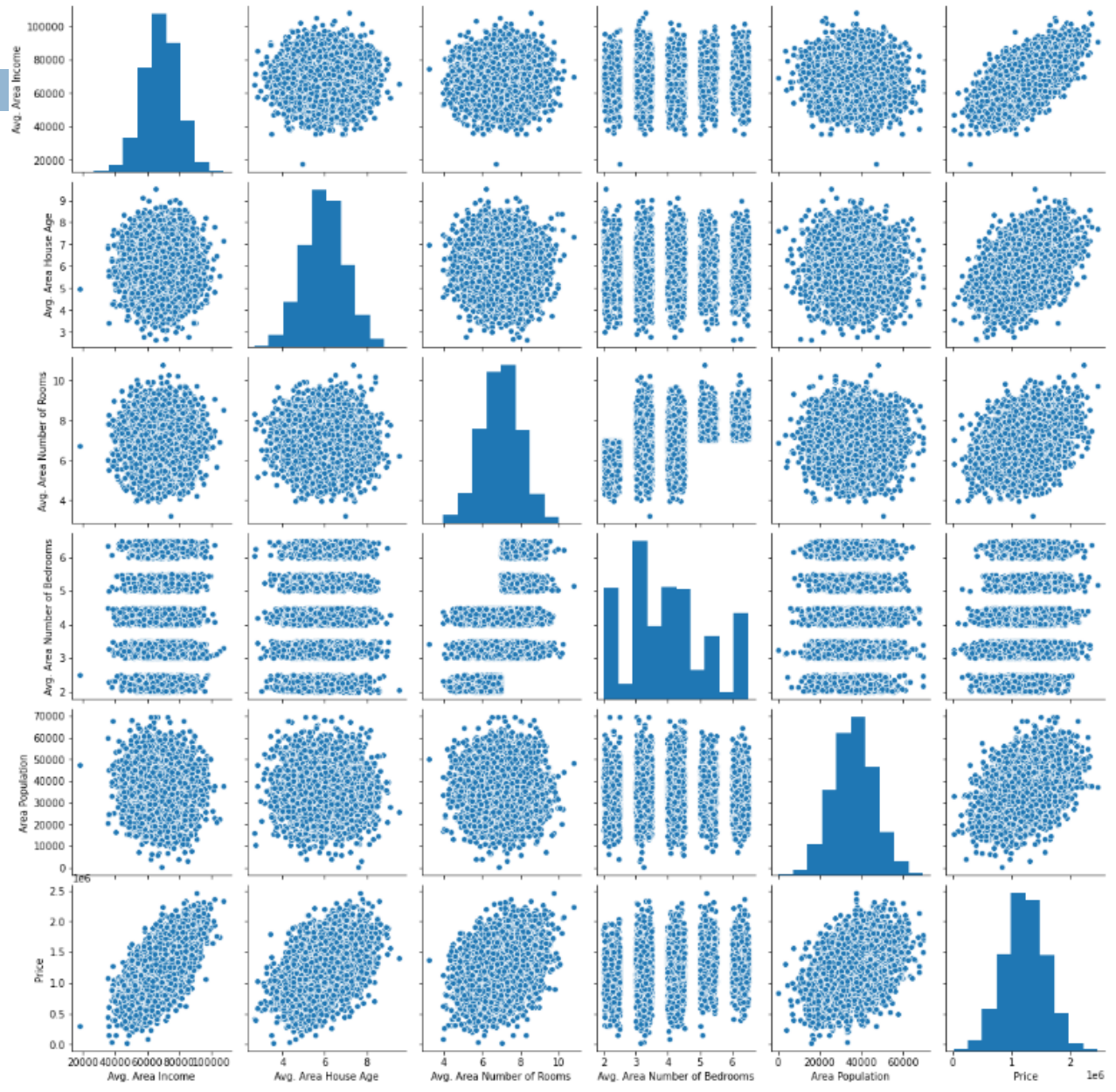

```
USAhousing.describe()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	9.975771e+05
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1.232669e+06
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1.471210e+06
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2.469066e+06

Exploratory Data Analysis

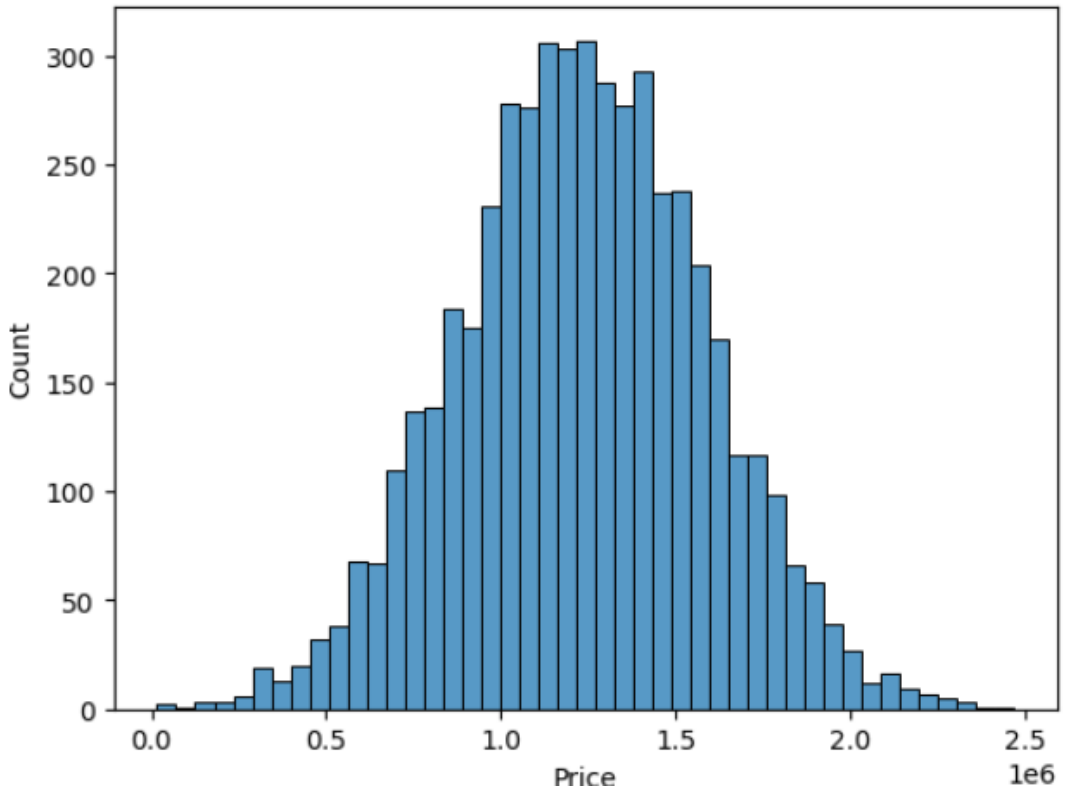
Plot the data to check it.

```
sns.pairplot(USAhousing, hue="Price")
```

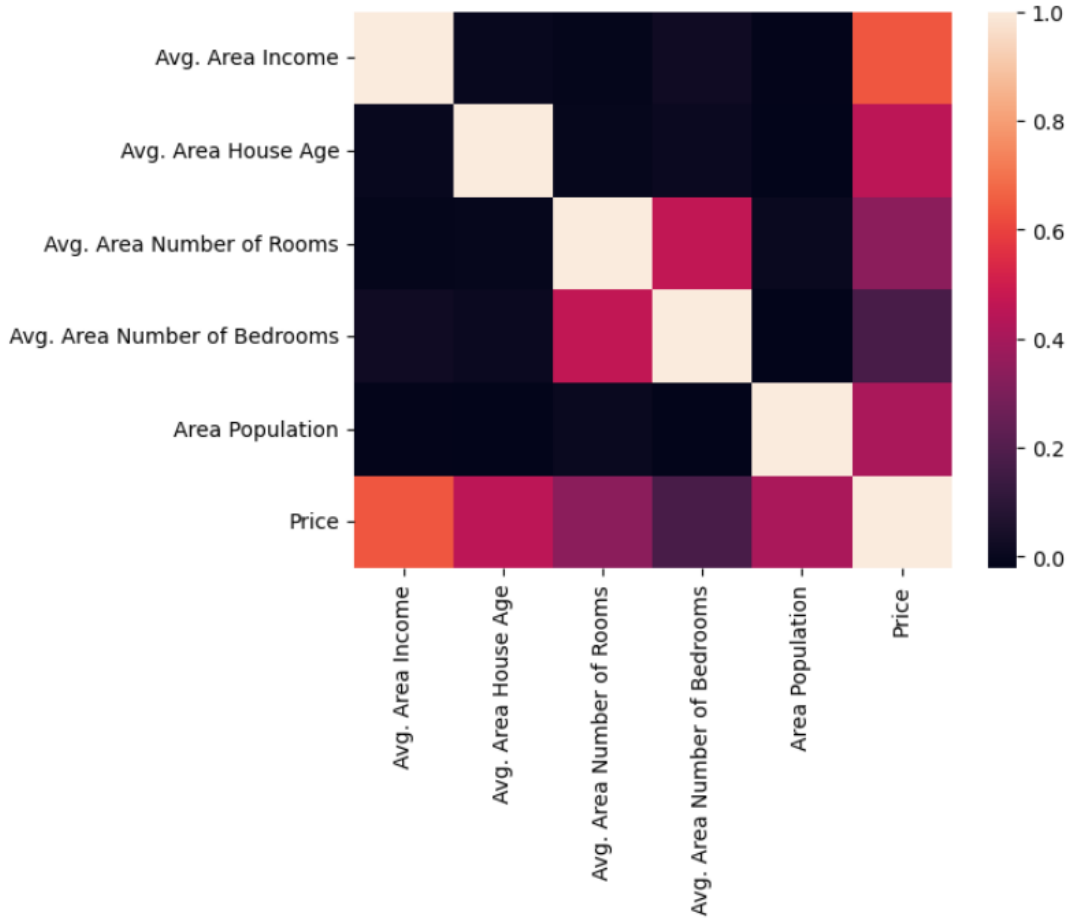


Exploratory Data Analysis


```
sns.histplot(USAhousing['Price'])
```




```
sns.heatmap(USAhousing.corr(numeric_only=True))
```



Linear Regression

The target is the ***Price***, so we will implement a Linear Regression model.
The feature *Address* will not be considered as it is neither relevant nor a numeric variable.

```
X = USAhousing[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
                'Avg. Area Number of Bedrooms', 'Area Population']]
y = USAhousing['Price']
```

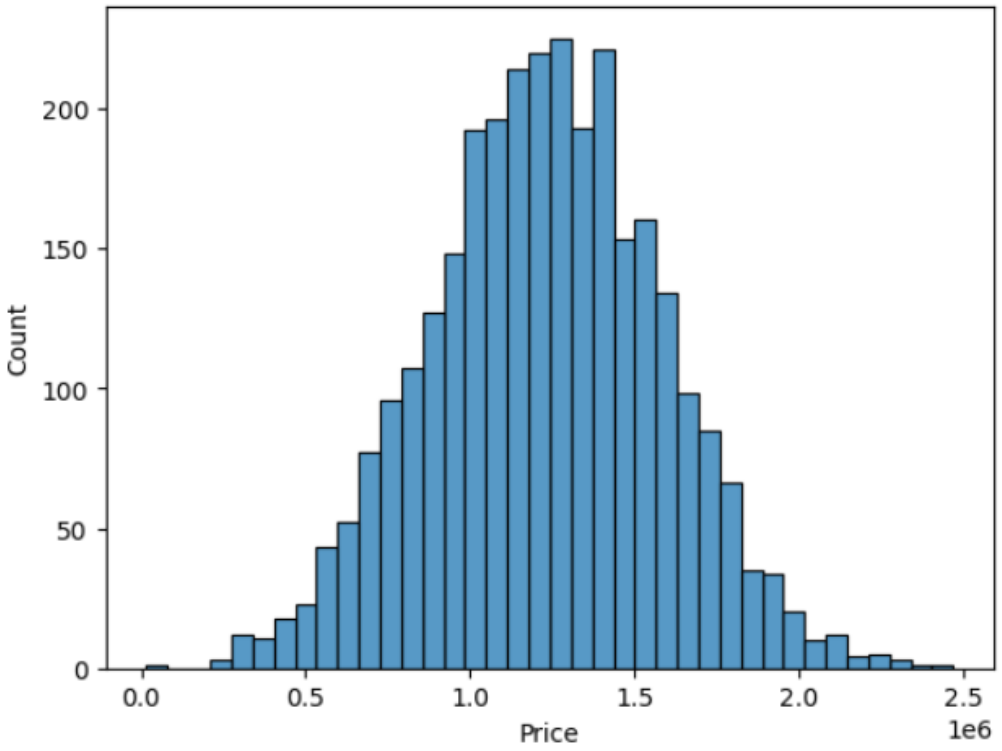
```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=101)
```

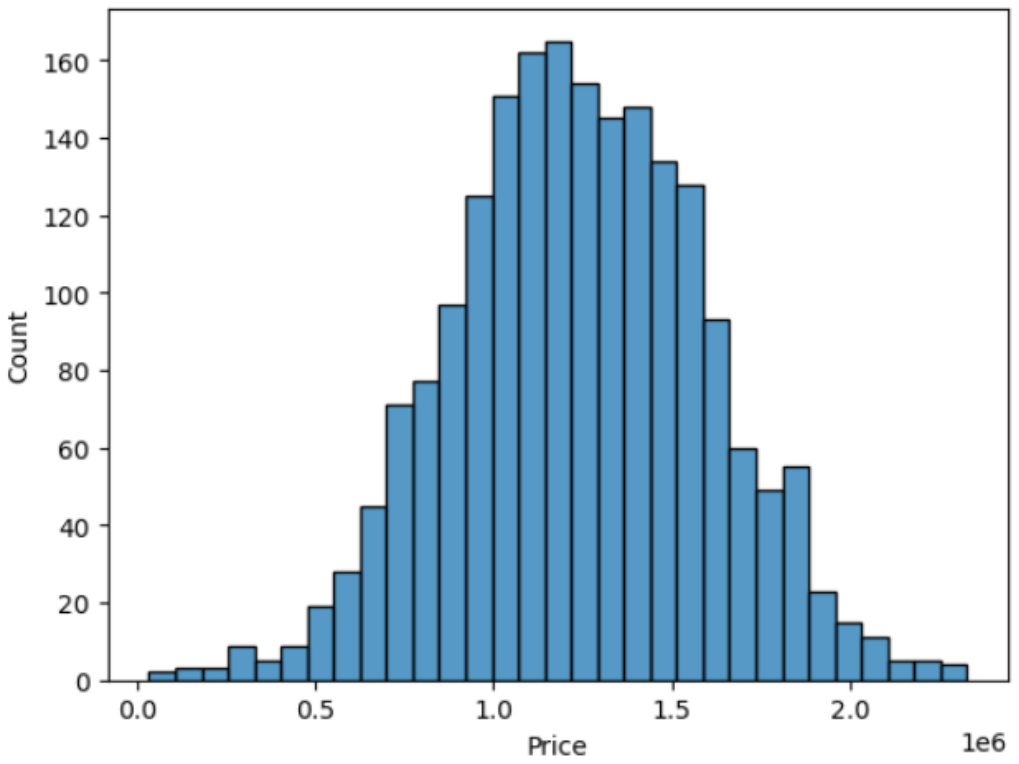
Linear Regression

Check the distribution of each subset:

```
sns.histplot(y_train)
```




```
sns.histplot(y_test)
```



Linear Regression


```
sklearn.linear_model.LinearRegression(*, fit_intercept=True, copy_X=True, n_jobs=None, positive=False)
```

Create and train the model:

```
from sklearn.linear_model import LinearRegression
```

```
lm = LinearRegression()
lm.fit(X_train,y_train)
```

```
▾ LinearRegression
LinearRegression()
```

Evaluate the model by checking and interpreting its coefficients:

```
print(lm.intercept_)
```

```
-2640159.7968525267
```

Linear Regression


```
coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

	Coefficient
Avg. Area Income	21.528276
Avg. Area House Age	164883.282027
Avg. Area Number of Rooms	122368.678027
Avg. Area Number of Bedrooms	2233.801864
Area Population	15.150420

Holding all the other features fixed:

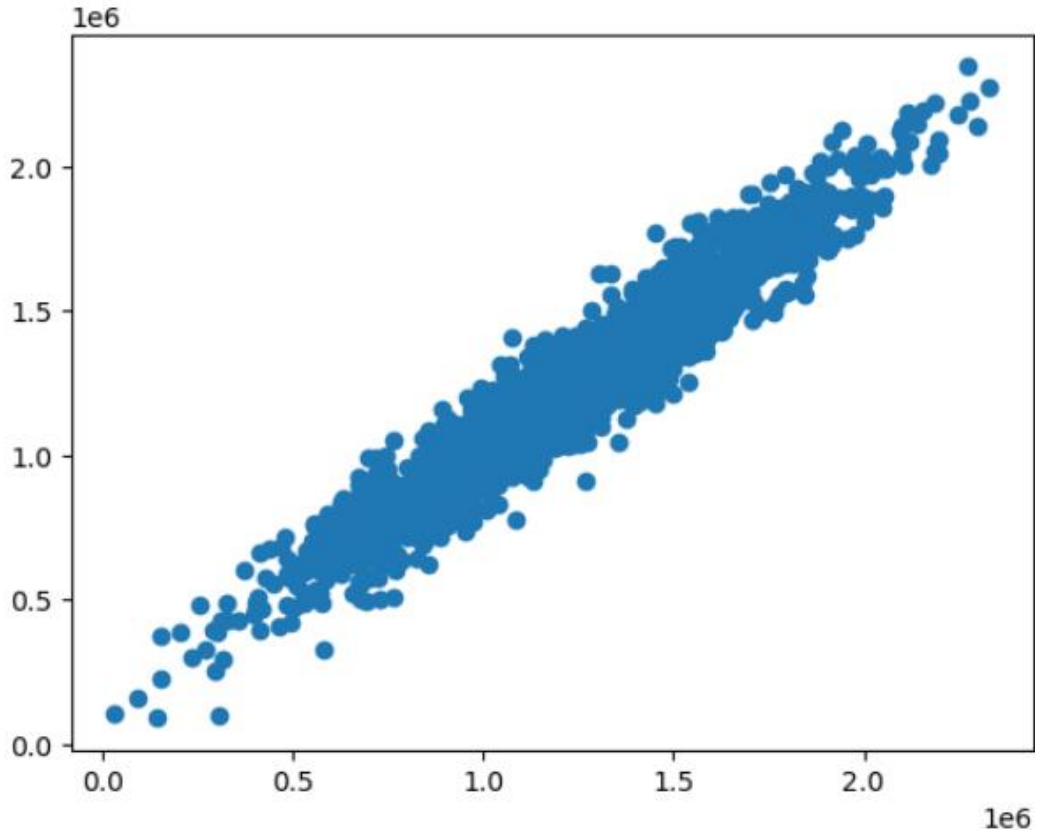
- An increase of 1 unit in ***Avg. Area Income*** is associated with an increase of $21.53
- An increase of 1 unit in ***Avg. Area House Age*** is associated with an increase of $164883.28
- An increase of 1 unit in ***Avg. Area Number of Rooms*** is associated with an increase of $122368.68
- An increase of 1 unit in ***Avg. Area Number of Bedrooms*** is associated with an increase of $2233.80
- An increase of 1 unit in ***Area Population*** is associated with an increase of $15.15

Linear Regression

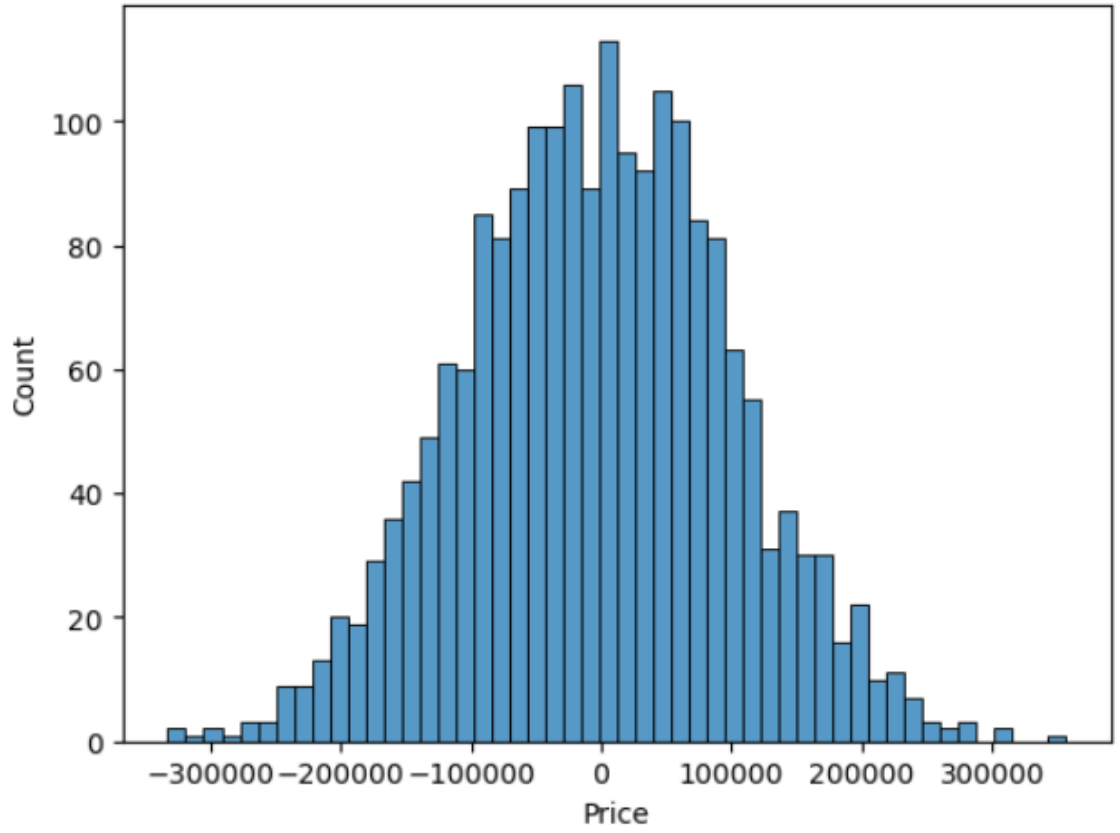
Analyze the predictions and plot it:

```
predictions = lm.predict(X_test)
```

```
plt.scatter(y_test,predictions)
```




```
sns.histplot((y_test-predictions),bins=50);
```



Linear Regression

Regression Evaluation Metrics

The three most common metrics used to evaluate regression problems are:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Comparing them:

- **MAE** is the easiest to understand because it is the average error;
- **MSE** is more popular than MAE because it "punishes" large errors;
- **RMSE** is even more popular than MSE because it is interpretable in units of the target variable.

All of these are **loss functions** because the aim is to minimize the error.

```
from sklearn import metrics
```

```
print('MAE:', metrics.mean_absolute_error(y_test, predictions))
print('MSE:', metrics.mean_squared_error(y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
MAE: 82288.22251914947
MSE: 10460958907.209057
RMSE: 102278.82922290935
```

Comparing Linear Regression with LASSO

Regularization

Technique of adding a penalty term to the objective function during training to prevent overfitting and improve generalization performance. The primary goal of regularization is to discourage the model from fitting the training data too closely, which can lead to poor performance on unseen data.

Generate random data and add it to new columns:

```
USAhousing['randomCol1'] = np.random.randint(1, 10, USAhousing.shape[0])
USAhousing['randomCol2'] = np.random.randint(10, 100, USAhousing.shape[0])
```

```
USAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 9 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object 
 7   randomCol1                    5000 non-null   int32  
 8   randomCol2                    5000 non-null   int32  
dtypes: float64(6), int32(2), object(1)
memory usage: 312.6+ KB
```

Comparing Linear Regression with LASSO


```
USAhousing.head()
```

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address	randomCol1	randomCol2
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...	7	46
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...	4	35
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482...	8	49
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820	5	89
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386	7	47

Comparing Linear Regression with LASSO


```
X = USAhousing[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
                'Avg. Area Number of Bedrooms', 'Area Population', 'randomCol1', 'randomCol2']]
y = USAhousing['Price']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=101)
```

Adjusting Linear Regression

```
lin_reg = LinearRegression()
lin_reg.fit(X, y)
```

LinearRegression

▼ Parameters

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

Comparing Linear Regression with LASSO


```
sklearn.linear_model.Lasso(alpha=1.0, *, fit_intercept=True, precompute=False, copy_X=True, max_iter=1000, tol=0.0001,
warm_start=False, positive=False, random_state=None, selection='cyclic')
```

```
from sklearn.linear_model import Lasso
```

Adjusting LASSO with regularization

```
lasso = Lasso(alpha=2000, random_state=2025)
lasso.fit(X, y)
```

Lasso

▼ Parameters

alpha	2000
fit_intercept	True
precompute	False
copy_X	True
max_iter	1000
tol	0.0001
warm_start	False
positive	False
random_state	2025
selection	'cyclic'

Comparing Linear Regression with LASSO

Comparing coefficients

```
results = pd.DataFrame({
    'Feature': [X.columns[i] for i in range(X.shape[1])],
    'Linear Regression': np.round(lin_reg.coef_, 2),
    'Lasso': np.round(lasso.coef_, 2)})
```

```
results
```

	Feature	Linear Regression	Lasso
0	Avg. Area Income	21.58	21.58
1	Avg. Area House Age	165790.06	163742.88
2	Avg. Area Number of Rooms	120768.29	119163.54
3	Avg. Area Number of Bedrooms	1652.12	949.99
4	Area Population	15.20	15.19
5	randomCol1	220.94	0.00
6	randomCol2	134.50	125.11

Using LASSO, we managed to **eliminate one coefficient** (*randomCol1*).
However, since we have very large coefficients and a small dataset, LASSO did not regularize well.

Logistic Regression

The Problem and the Data

Problem: development of a Machine Learning model that predicts **which passengers survived** the Titanic shipwreck.

Approach: **Logistic Regression** approach.

Dataset: information about the passengers, including:

- ***survival***: whether the passenger survived (0: No, 1: Yes);
- ***pclass***: ticket class (1: 1st, 2: 2nd, 3: 3rd);
- ***sex***: M: Male, F: Female;
- ***Age***: age in years;
- ***sibsp***: number of siblings per spouse aboard;
- ***parch***: number pf parents per children aboard;
- ***ticket***: ticket number;
- ***fare:*** passenger fare;
- ***cabin:*** cabin number;
- ***embarked:*** port of embarkation (C: Cherbourg, Q: Queenstown, S: Southampton).

Exploratory Data Analysis


```
train = pd.read_csv('titanic_train.csv')
```

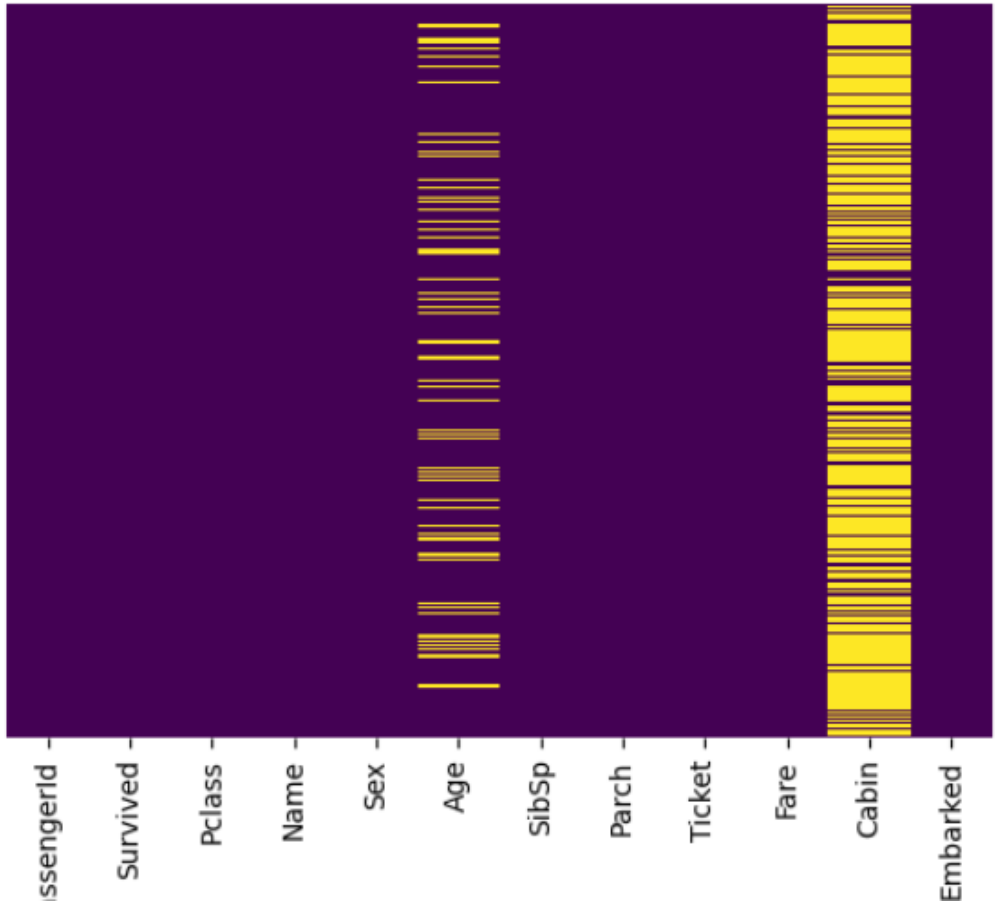
```
train.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Exploratory Data Analysis

Check for missing data:

```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```



Approximately **20%** of the *Age* data is **missing**. This proportion is likely small enough to be reasonably replaced with some form of **imputation**.

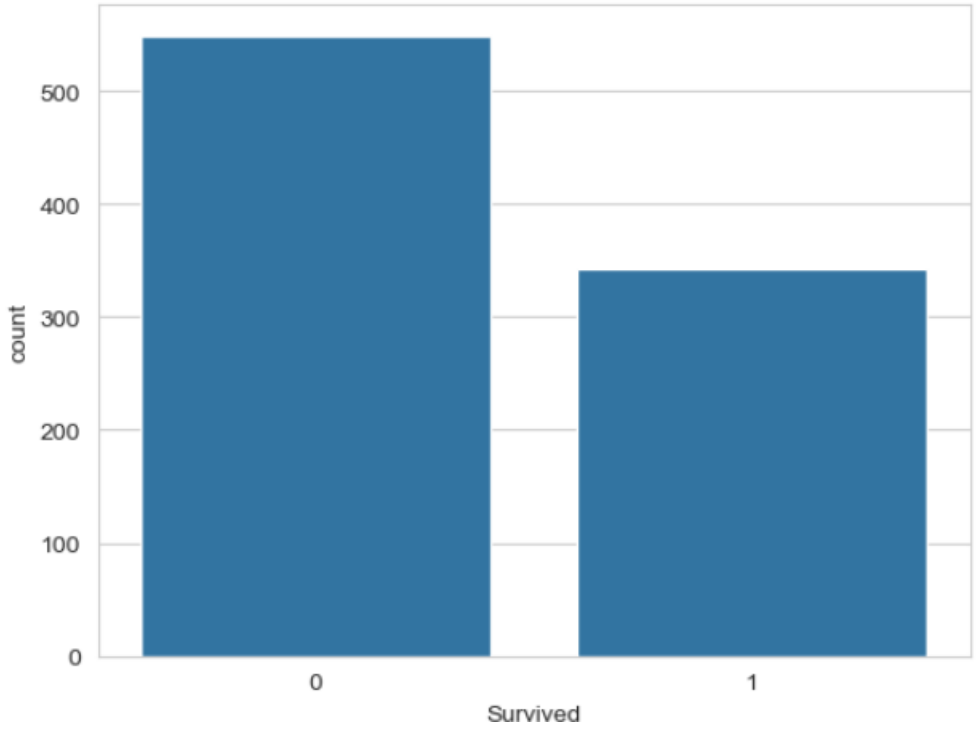
However, looking at the *Cabin* column, it appears that we are **missing too much** of that data to do anything useful with it at a basic level. We will probably remove this feature later or replace it with another feature, such as "Cabin Known: 1 or 0".

Exploratory Data Analysis

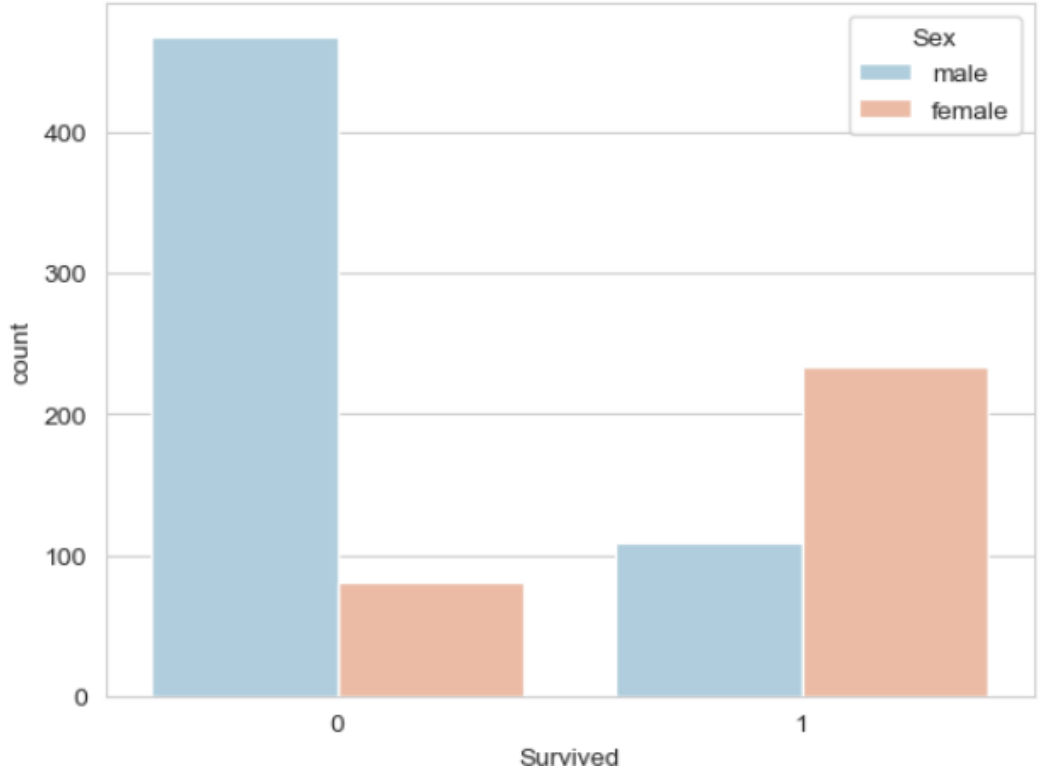
Continue by visualizing more of the data.

Survived

```
sns.set_style('whitegrid')
sns.countplot(x='Survived', data=train)
```



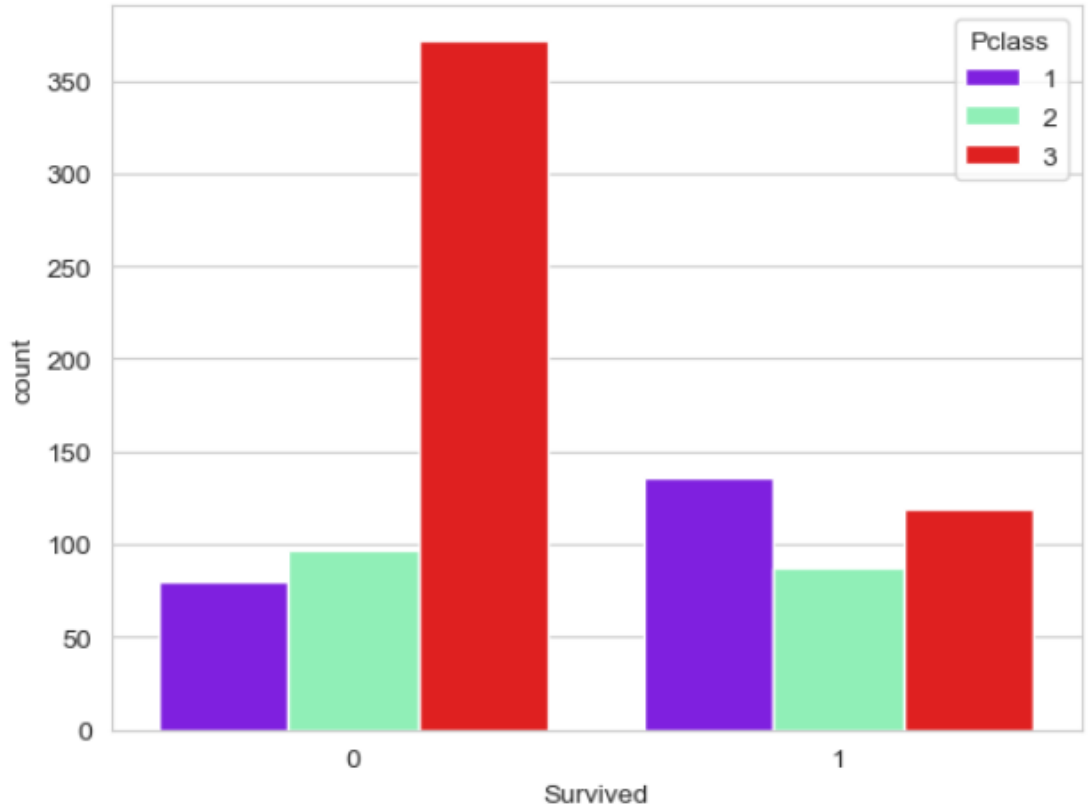

```
sns.set_style('whitegrid')
sns.countplot(x='Survived',hue='Sex',data=train,palette='RdBu_r')
```



by ***Sex***

Exploratory Data Analysis

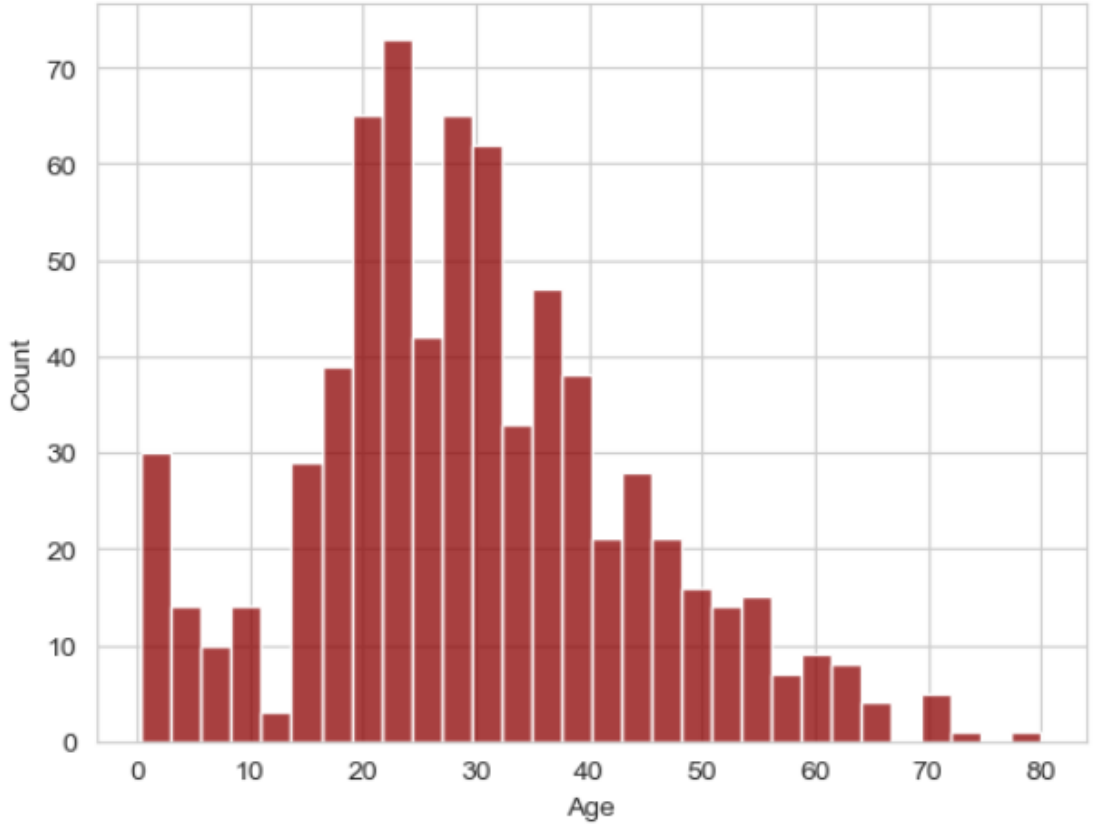

```
sns.set_style('whitegrid')
sns.countplot(x='Survived',hue='Pclass',data=train,palette='rainbow')
```



by ***Pclass***

Age

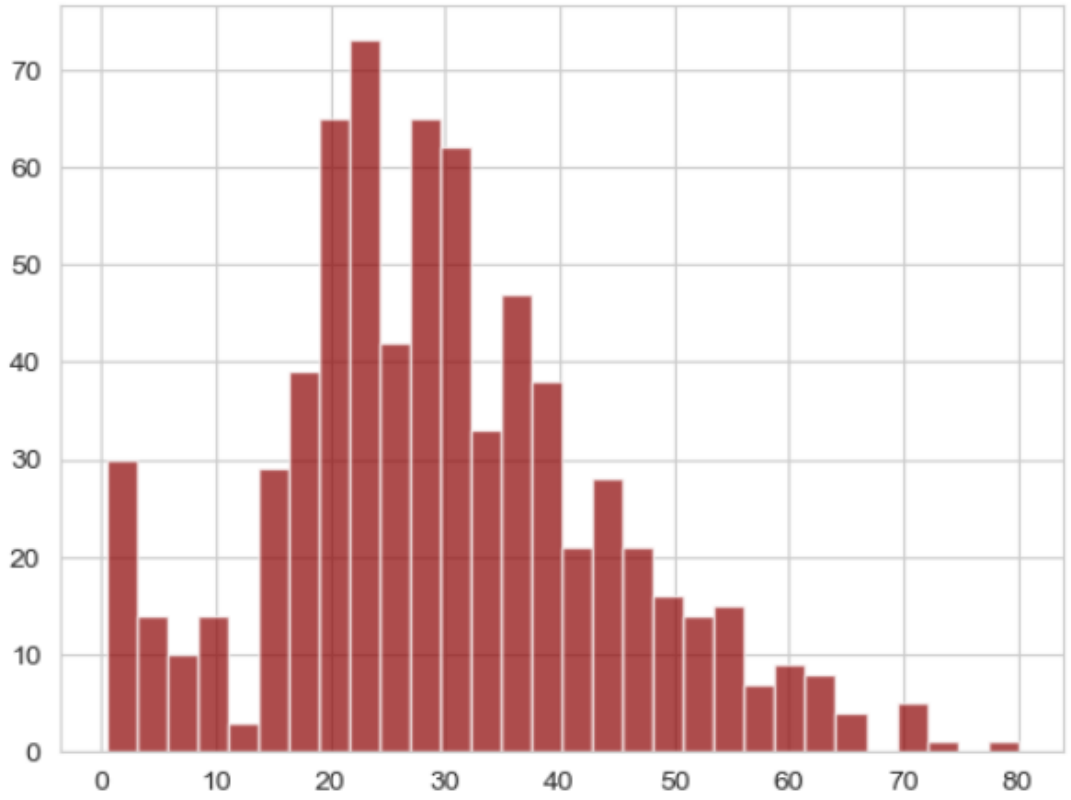
```
sns.histplot(train['Age'].dropna(),kde=False,color='darkred',bins=30)
```



Exploratory Data Analysis

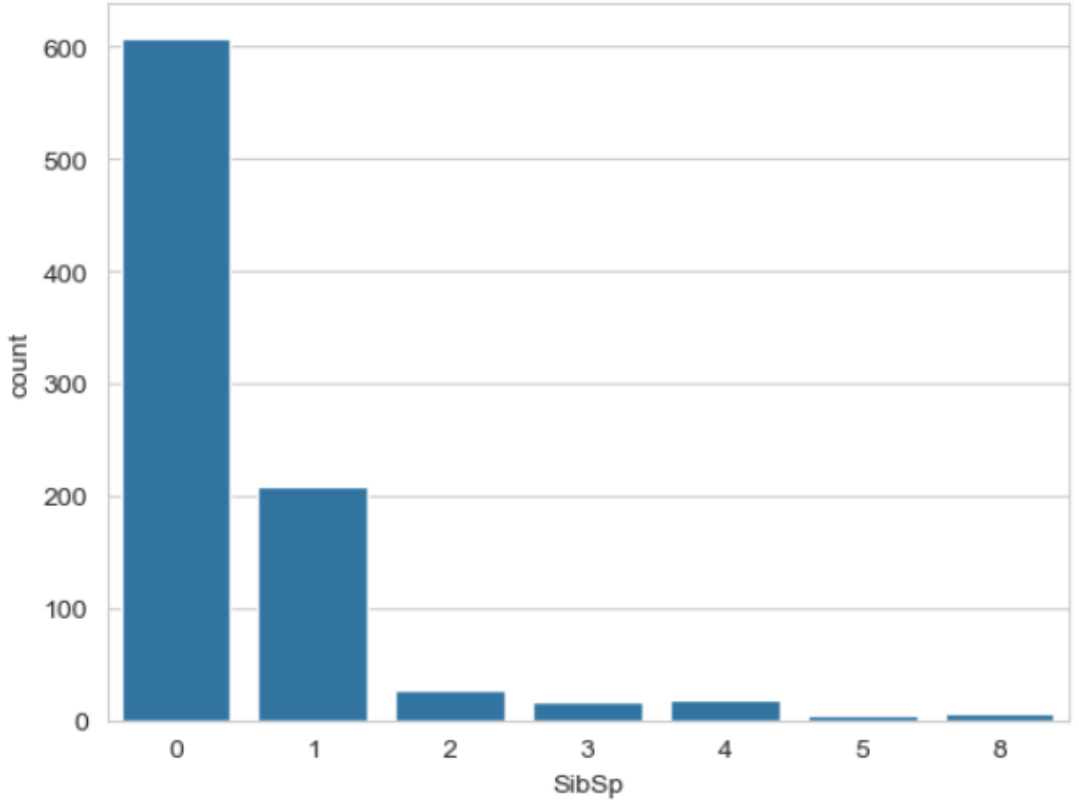
Age

```
train['Age'].hist(bins=30,color='darkred',alpha=0.7)
```



SibSp

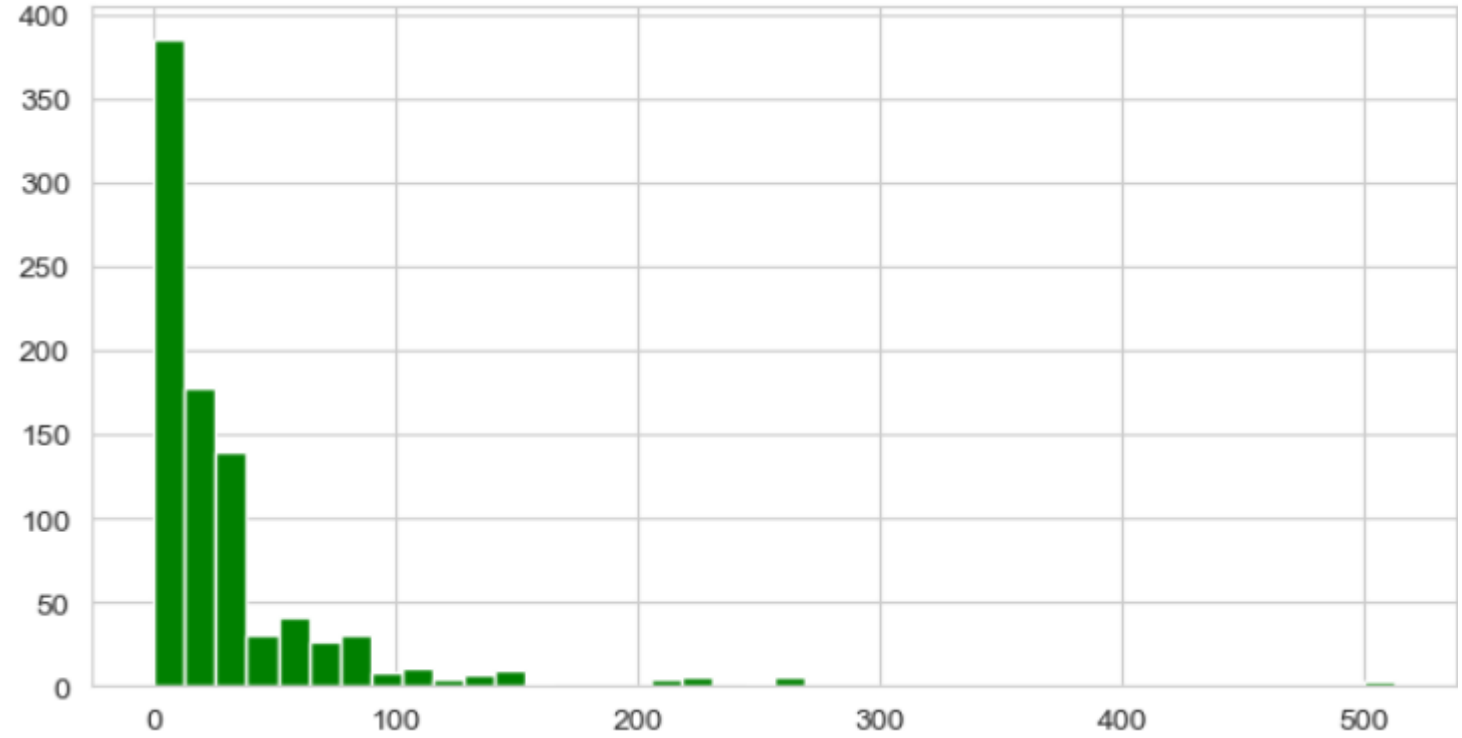
```
sns.countplot(x='SibSp',data=train)
```



Exploratory Data Analysis

Fare

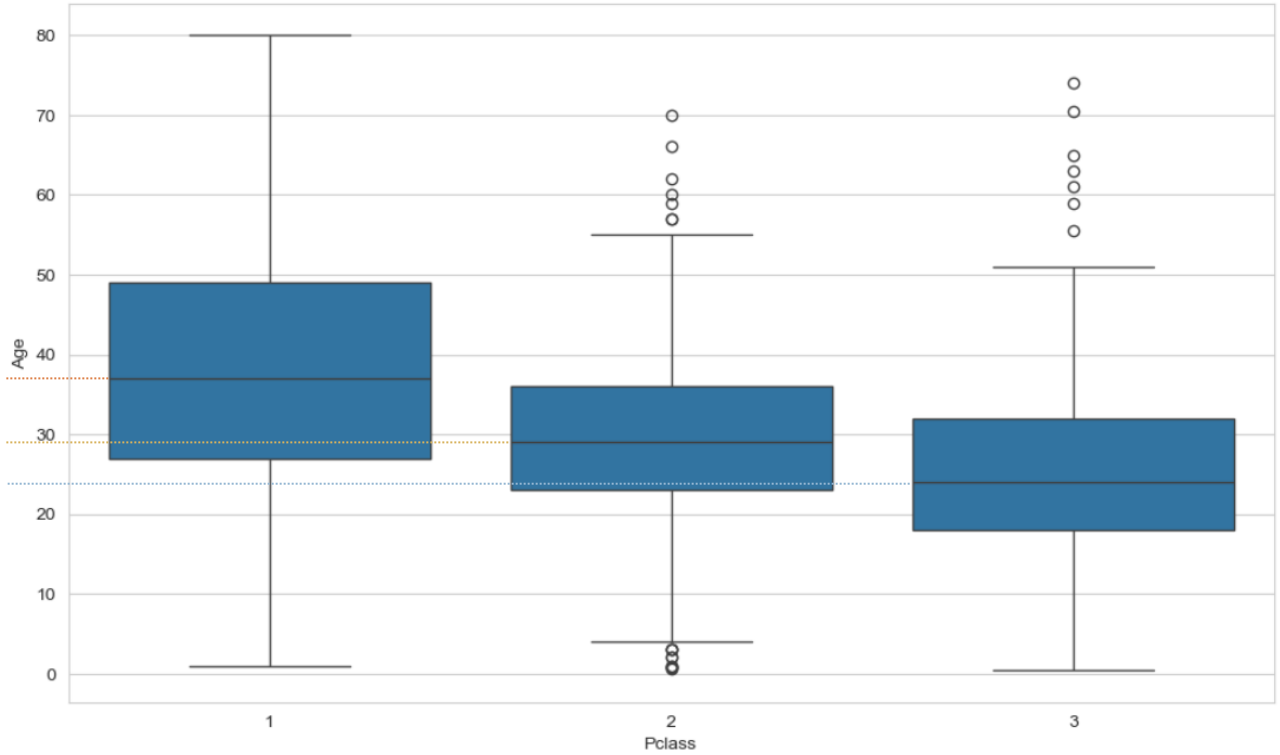
```
train['Fare'].hist(color='green',bins=40,figsize=(8,4))
```



Data Preprocessing

Instead of removing it, we want to **fill in missing values**. One way to do this is to impute the average age of all passengers (**imputation**).

```
plt.figure(figsize=(12, 7))
sns.boxplot(x='Pclass', y='Age', data=train)
```



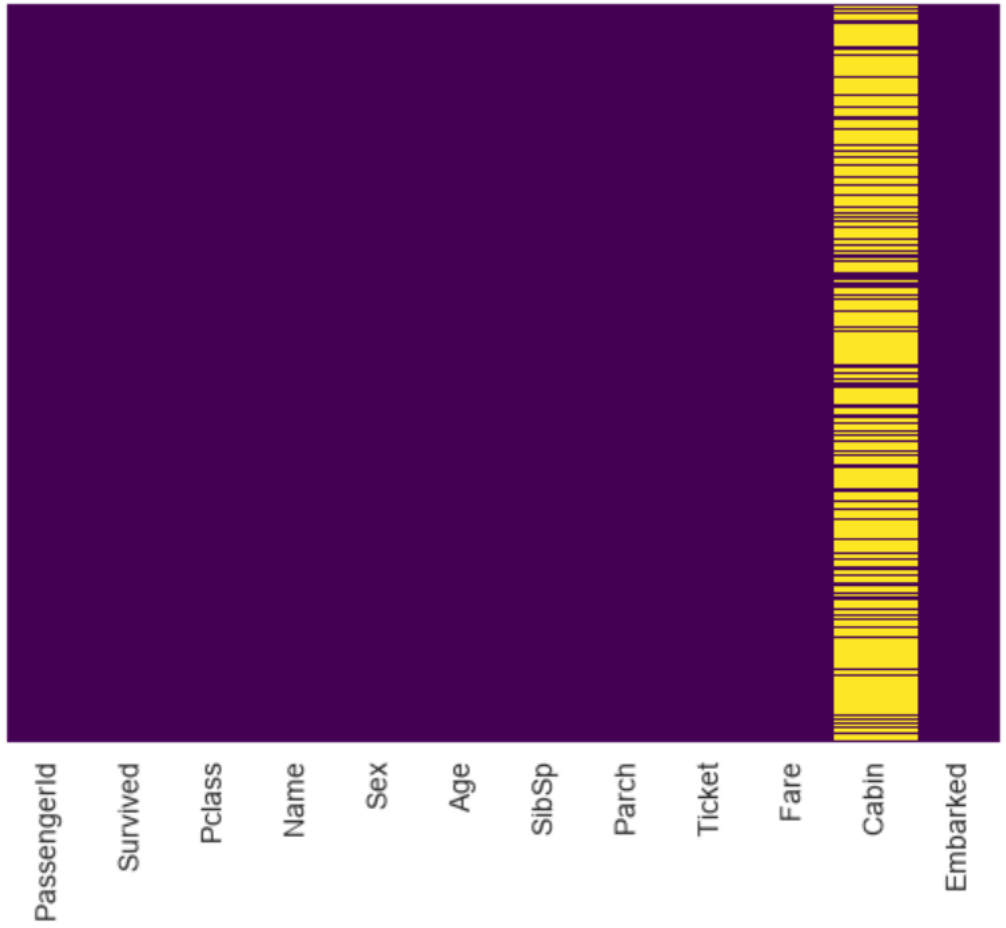
As expected, we can see that wealthier passengers in the higher classes tend to be older.

Data Preprocessing

We will use the **median age values** to impute based on *Pclass*.

```
train['Age'] = train['Age'].fillna(train.groupby('Pclass')['Age'].transform('median'))
```

```
sns.heatmap(train.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```



Data Preprocessing

Remove the *Cabin* column and the *Embarked* rows that are *NaN*:

```
train.drop('Cabin',axis=1,inplace=True)
```

```
train.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

```
train.dropna(inplace=True)
```

Data Preprocessing

Use the Pandas library to convert categorical features into dummy variables:

```
train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 889 entries, 0 to 890
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  889 non-null    int64
 1   Survived     889 non-null    int64
 2   Pclass       889 non-null    int64
 3   Name         889 non-null    object
 4   Sex          889 non-null    object
 5   Age          889 non-null    float64
 6   SibSp        889 non-null    int64
 7   Parch        889 non-null    int64
 8   Ticket       889 non-null    object
 9   Fare         889 non-null    float64
 10  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(4)
memory usage: 83.3+ KB
```

Sex, Embarked

```
sex = pd.get_dummies(train['Sex'],drop_first=True)
embark = pd.get_dummies(train['Embarked'],drop_first=True)
```

```
train.drop(['Sex','Embarked','Name','Ticket'],axis=1,inplace=True)
```

```
train = pd.concat([train,sex,embark],axis=1)
```

```
train.head()
```

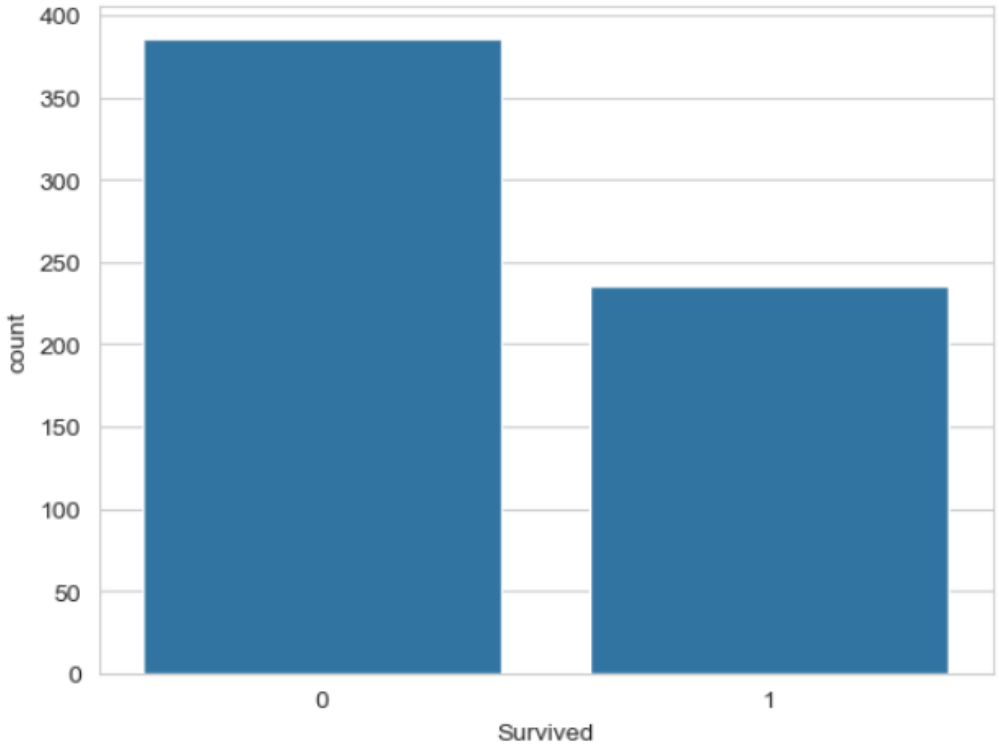
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	male	Q	S
0	1	0	3	22.0	1	0	7.2500	True	False	True
1	2	1	1	38.0	1	0	71.2833	False	False	False
2	3	1	3	26.0	0	0	7.9250	False	False	True
3	4	1	1	35.0	1	0	53.1000	False	False	True
4	5	0	3	35.0	0	0	8.0500	True	False	True

Logistic Regression

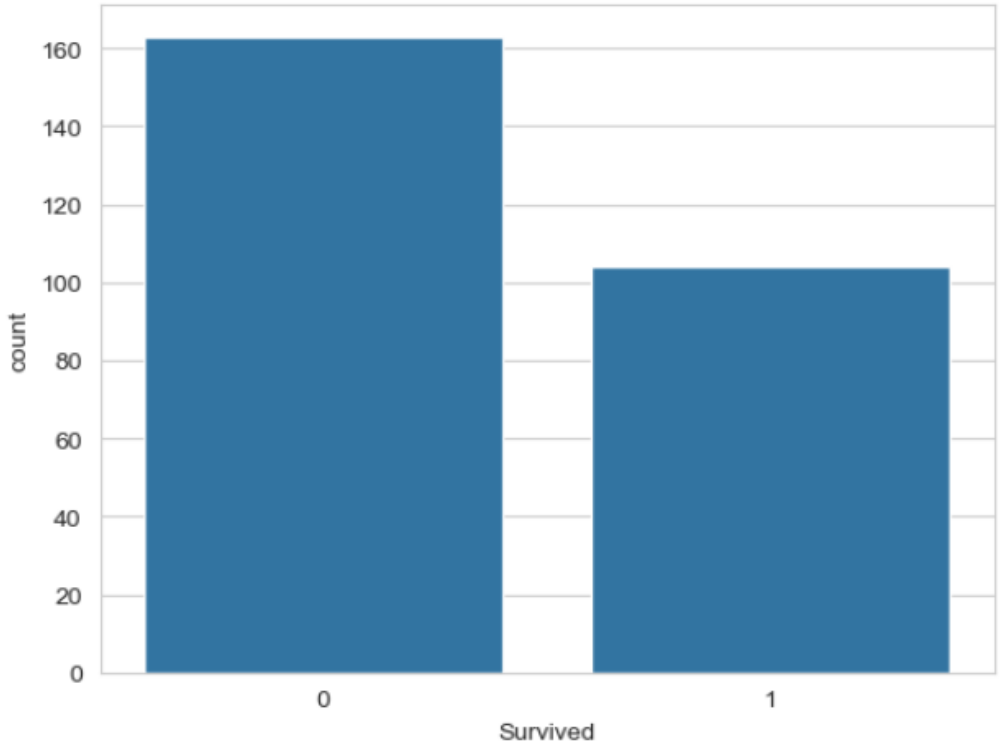

```
from sklearn.model_selection import train_test_split
X = train.drop('Survived',axis=1)
y = train['Survived']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30, random_state=101)
```

```
sns.set_style('whitegrid')
sns.countplot(x='Survived', data = pd.DataFrame(y_train,columns=['Survived']) )
```




```
sns.set_style('whitegrid')
sns.countplot(x='Survived', data = pd.DataFrame(y_test,columns=['Survived']) )
```



Logistic Regression


```
sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False, tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None,
                   random_state=None, solver='lbfgs', max_iter=100, multi_class='auto', verbose=0, warm_start=False, n_jobs=None, l1_ratio=None)
```

Logistic Regression **solvers**:

- For small datasets, *liblinear* is a good choice, whereas *sag* and *saga* are faster for larger ones;
- Only *newton-cg*, *sag*, *saga* and *lbfgs* handle multinomial loss for multiclass problems;
- *liblinear* is limited to one-versus-rest schemes.

Supported **penalties** by solver:

- ***newton-cg*** – *[l2, none]*
- ***lbfgs*** – *[l2, none]*
- ***liblinear*** – *[l1, l2]*
- ***sag*** – *[l2, none]*
- ***saga*** – *[elasticnet, l1, l2, none]*

```
from sklearn.linear_model import LogisticRegression
```

Logistic Regression

Model 1: random_state = 2022, solver = 'newton-cg'

```
starttime = time.process_time()

logmodel1 = LogisticRegression(random_state=2022, solver='newton-cg')
print(logmodel1)
logmodel1.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")
```

```
LogisticRegression(random_state=2022, solver='newton-cg')
Time spent: 0.015625 seconds
```

```
predictions1 = logmodel1.predict(X_test)
```

Model 2: random_state = 2022, solver = 'lbfgs'

```
starttime = time.process_time()

logmodel2 = LogisticRegression(random_state=2022, solver='lbfgs', max_iter=800)
print(logmodel2)
logmodel2.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")
```

```
LogisticRegression(max_iter=800, random_state=2022)
Time spent: 0.109375 seconds
```

```
predictions2 = logmodel2.predict(X_test)
```

Model 3: random_state = 2022, solver = 'liblinear'

```
starttime = time.process_time()

logmodel3 = LogisticRegression(random_state=2022, solver='liblinear')
print(logmodel3)
logmodel3.fit(X_train,y_train)

endtime = time.process_time()
print(f"Time spent: {endtime - starttime} seconds")
```

```
LogisticRegression(random_state=2022, solver='liblinear')
Time spent: 0.0 seconds
```

```
predictions3 = logmodel3.predict(X_test)
```

Logistic Regression

Use the following metrics to evaluate the model: *precision*, *recall*, *F1-score* and *confusion matrix*

```
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
```

```
print("With 'newton-cg': \n", classification_report(y_test,predictions1))
print("With 'lbfgs': \n", classification_report(y_test,predictions2))
print("With 'liblinear': \n", classification_report(y_test,predictions3))
```

With 'newton-cg':

	precision	recall	f1-score	support
0	0.82	0.91	0.86	163
1	0.84	0.68	0.75	104
accuracy			0.82	267
macro avg	0.83	0.80	0.81	267
weighted avg	0.83	0.82	0.82	267

With 'lbfgs':

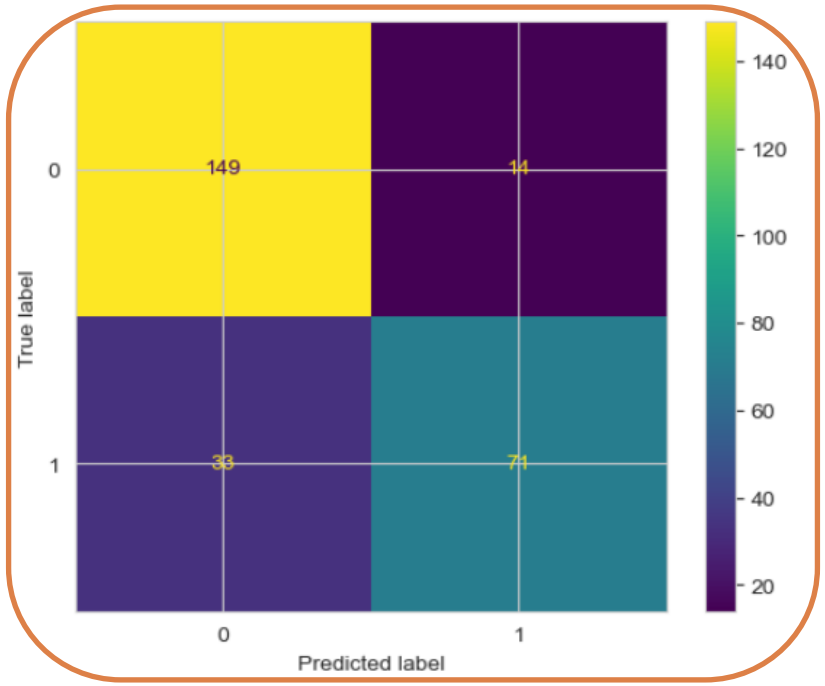
	precision	recall	f1-score	support
0	0.82	0.91	0.86	163
1	0.84	0.68	0.75	104
accuracy			0.82	267
macro avg	0.83	0.80	0.81	267
weighted avg	0.83	0.82	0.82	267

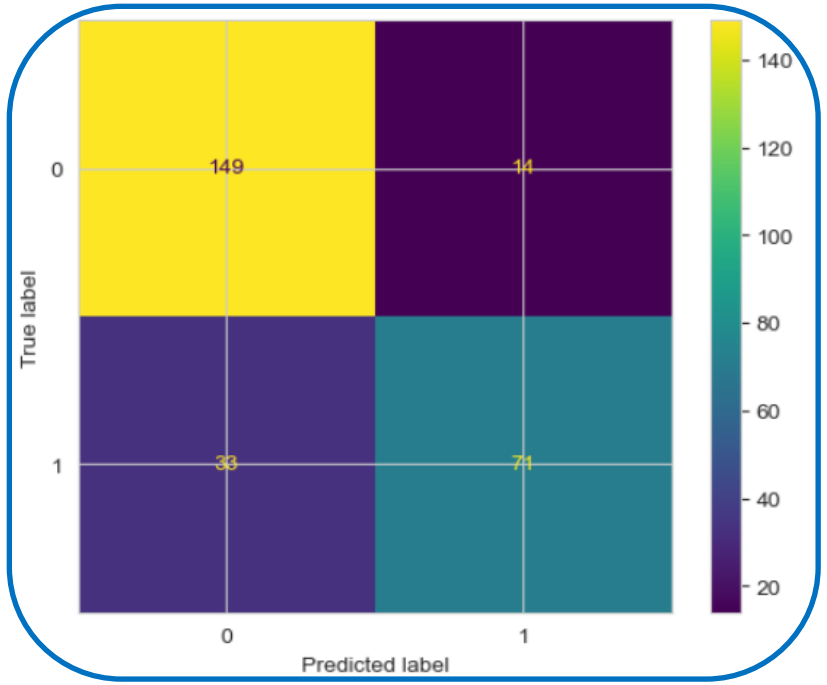
With 'liblinear':

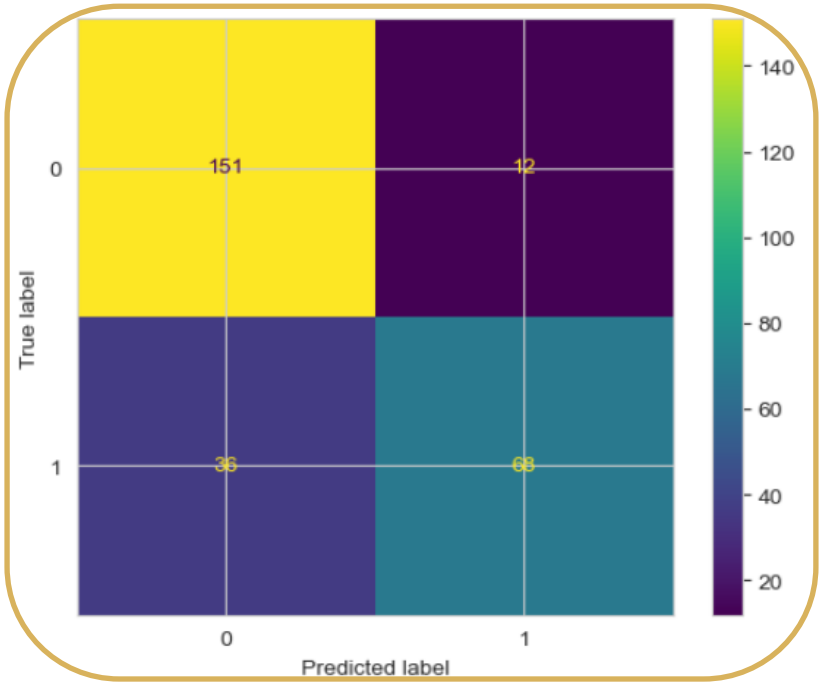
	precision	recall	f1-score	support
0	0.81	0.93	0.86	163
1	0.85	0.65	0.74	104
accuracy			0.82	267
macro avg	0.83	0.79	0.80	267
weighted avg	0.82	0.82	0.81	267

Logistic Regression


```
ConfusionMatrixDisplay.from_predictions(y_test, predictions1)
ConfusionMatrixDisplay.from_predictions(y_test, predictions2)
ConfusionMatrixDisplay.from_predictions(y_test, predictions3)
plt.show()
```







Hands On